HI 744 | Patient Similarity Retrieval

Programming Assignment — Fall 2026 | Technical Report

Objective and dataset

Build a TF-IDF baseline for clinical-note similarity, compare it with Okapi BM25, retrieve five other patients for every record, and evaluate precision@5 and recall@5. The provided PMC-Patients subset contains **8,579** JSON records, averaging **451.9 words** per clinical case. Each record supplies patient_uid, patient_id, patient text and similar_patients relevance annotations. No sampling was used.

1. Data preparation

Records are read from JSON files and sorted by path. Clinical text is lowercased and tokenized, punctuation and English stopwords are removed, and selected plural suffixes are normalized by conservative rules. This last step is only an approximation to lemmatization, not a complete linguistic or biomedical lemmatizer. scikit-learn CountVectorizer constructs a sparse count matrix with at most 30,000 features.

2. Retrieval methods

TF-IDF: scikit-learn TfidfTransformer applies smoothed IDF, sublinear term frequency (1 + log(tf)) and L2 normalization. The cosine similarity between two normalized vectors equals their dot product. Smoothing stabilizes IDF, logarithmic TF reduces repeated-term dominance, and L2 normalization limits the effect of record length.

BM25: Okapi BM25 uses k1=1.5 and b=0.75. It saturates term frequency and explicitly adjusts for document length. Every distinct term in a patient record is treated as a query term, scored against all other records. Both methods are lexical and may miss clinically equivalent expressions that share few words.

For both retrieval methods, the patient is excluded from its own results; the top five are ranked by descending similarity/score, with corpus-index tie-breaking.

3. Evaluation

Relevance comes from similar_patients: both grade 1 (moderate) and grade 2 (high) count equally. **Precision@5 = relevant retrieved / 5. Recall@5 = relevant retrieved / all annotated relevant patients.** For queries with no annotated relevant patients, recall is stored as null and omitted from the mean; precision is still calculated. Annotated patients absent from the supplied corpus remain in the recall denominator, as specified by the assignment.

4. Results

Method	Mean precision@5	Mean recall@5	Recall excluded
TF-IDF	0.006598	0.028981	3,777
BM25	0.005572	0.024677	3,777

TF-IDF performed slightly better on this subset. Its average precision@5 of 0.00660 means roughly 0.033 labeled relevant patients per five-result list; BM25 yielded about 0.028. For queries with at least one annotated match, TF-IDF recovered about 2.90% of annotated matches, versus 2.47% for BM25.

The methods' mean top-five overlap is **0.5589** (55.89%), equivalent to about 2.79 shared results out of five per patient. Thus, they agree on a little over half of the retrieved neighbors.

5. Interpretation and limitations

The low scores should not be interpreted as a validated measure of clinical utility. There are 10,950 annotated relevance links to patients outside the supplied candidate pool, reducing attainable recall. Lexical overlap, variable note length and vocabulary, and incomplete annotations also affect the evaluation. Neither method is appropriate for clinical decision support without further validation.

6. Reproducibility

From the project root, create and activate a Python environment, install dependencies using **python -m pip install -r requirements.txt**, then execute **python patient_similarity_retrieval.py data/PMC_Patients**. Optional flags: --output-dir and --max-features. The project uses separate src modules for loading, preprocessing, vectorization, ranking, BM25, evaluation and orchestration.

7. Submitted outputs

Task 1: outputs/tfidf_matrix.npz is a SciPy sparse matrix. outputs/tfidf_metadata.json supplies patient_uid-to-row and vocabulary-term-to-column mappings, enabling direct weight lookup by patient_uid and term.

Task 2: outputs/patient_similarity.json maps each patient_uid to patient_id and ranked top-five patient_uid lists for tfidf and bm25.

Task 3: outputs/performance.json maps each patient_uid to precision_5 and recall_5 for both methods. outputs/summary.json contains the aggregate scores and dataset statistics.

8. Future improvements

Use a full biomedical lemmatizer with clinically appropriate exception handling, compare against biomedical embedding models, and evaluate with a candidate-complete corpus. A separate analysis by condition and relevance grade could reveal important failure modes that mean precision and recall obscure.

Source

HI 744 Fall 2026 Programming Assignment; supplied PMC_Patients.zip. Dataset documentation: zhengyun21/PMC-Patients (Hugging Face).